

Antonio Loquercio

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Last Update: July 31, 2026

Education

- Feb 17–July 21 **Ph.D.** at the *University of Zurich*. Honors: *Summa cum laude*
Dissertation title: *Learning Agile Robot Navigation*.
Dissertation Committee: Prof. Peter Abbeel (UC Berkeley), Prof. Angela Schoellig (U. of Toronto), Prof. Roland Siegwart (ETH)
Advisor: Prof. Davide Scaramuzza
- Sep 14–Jan 17 **M.Sc.** in Robotics, System and Control at *ETH Zurich*. Honors: *Summa cum laude*
Overall Grade Point Average: 5.92 (out of 6)
Master Thesis: *Efficient Descriptor Learning for Large Scale Localization*
Advisor: Prof. Roland Siegwart
- Sep 11–July 14 **B.Sc.** in Advanced Control and Informatics, *Università Tor Vergata, Rome*
Bachelor Thesis: The Unscented and Extended Kalman Filter in Mobile Robotics
Advisor: Prof. Francesco Martinelli

Awards

- 2025 **Mario Gerla Young Investigator Award**
- 2023 **Article featured on the cover of *Nature***
(Paper: Champion-level Drone Racing with Deep Reinforcement Learning)
- 2023 **Frontiers of Science Award**
(Paper: Learning High-Speed Flight in the Wild, Science Robotics)
- 2022 **Georges Giralt Ph.D. Award**, the most prestigious recognition
for a PhD dissertation in robotics in Europe.
- 2022 **Outstanding Reviewer Award**, Robotics and Automation Letters (RA-L).
- 2020 **Best Paper Award** Honorable Mention, IEEE Transactions on Robotics (T-RO)
(paper: Deep Drone Racing: From Simulation to Reality with Domain Randomization)
- 2020 **Best Paper Award** Finalist, Robotics, Science and Systems (RSS)
(paper: Deep Drone Acrobatics)

- 2018 **Best System Paper Award**, Conference on Robotics Learning (CORL)
(paper: Deep Drone Racing: Learning Agile Flight in Dynamic Environments)
- 2017 **ETH Medal** for outstanding master thesis
(awarded to each department's best master theses at ETH Zurich)

Experience

- Nov 21–Jun 24 **Postdoctoral scholar** at *University of California at Berkeley*, advised by Jitendra Malik
- Feb 17–Sep 21 **Graduate Student Researcher** at *University of Zurich* and *ETH Zurich*
at *Robotics and Perception Group*, advised by Davide Scaramuzza
- Feb 18–Sept 18 **Graduate Student Researcher** at *University of California Los Angeles*
at *UCLA Vision Lab*, advised by Stefano Soatto

Fellowships

- 2015 Excellence Scholarship and Opportunity Program, ETH Zurich
(ETH most prestigious scholarship award for master students)
- 2012–2013 Merit Scholarship Faculty of Engineering, Università Roma Tor Vergata
(awarded annually to the best student of the faculty of Engineering)
- 2012–2013 Top Ten Students in Engineering Sciences, Università Roma Tor Vergata
(awarded annually to the top 10 students of the Engineering Sciences study program)
- 2011–2013 Collegio Universitario Lamaro Pozzani Scholarship
(national scholarship covering all living and study costs for university students)
- 2011 Rotary Club Merit Scholarship
(awarded to the top five high-school graduates in Viterbo, Italy)

Educational Activities

- 2026 Spring. **ESE651**, *Physical Intelligence*, UPenn
- 2025 Fall. **ESE651**, *Physical Intelligence*, UPenn
- 2025 Spring. **ESE68** at *Real-World Robot Learning*, UPenn
- 2022 **Lecturer** at *Materials+*, *The AI PowerBoat Project*, ETH Zurich
- 2021 Guest Lecturer at *Aerial robotics*, EPFL Lausanne
- 2021 **Lecturer** at *Materials+*, *The AI PowerBoat Project*, ETH Zurich
- 2020 Guest Lecturer at *Vision Algorithms for Mobile Robotics*, ETH Zurich
- 2020 Guest Lecturer at *DSI Studium Digitale*, University of Zurich
- 2019 Guest Lecturer at *Vision Algorithms for Mobile Robotics*, ETH Zurich
- 2017–2018 Teaching Assistant at *Vision Algorithms for Mobile Robotics*, ETH Zurich
- 2016 Teaching Assistant at *Advanced Machine Learning*, ETH Zurich

Media Coverage

22.04.2026	Our paper on multi-agent learning is featured in NSF ACCESS .
10.10.2024	Our paper on learning adaptive control is featured in IEEE Spectrum .
31.09.2023	My Nature paper on beating the best human pilots in drone racing has received great media attention: [IEEE Spectrum , SRF , TeleZürich , The Guardian , The Daily Mail , Spiegel , Heise , National Public Radio , The New Scientist , El Diario , NZZ , Forbes]
26.10.2021	Forbes , This hotshot AI drone can speed through complex environments thanks to new kind of virtual training [Link]
07.10.2021	IEEE Spectrum , Autonomous Racing Drones Dodge Through Forests at 40 kph [Link]
07.10.2021	Robohub , Flying high-speed drones into the unknown with AI [Link]
29.06.2020	Der Spiegel , Akrobatische Drohnen [PDF]
25.06.2020	Drones Crunch , Must Watch! Programming Precision Aerobatics [Link]
24.06.2020	NCYT , Acrobacias para drones [Link]
24.06.2020	ZDNet , An autonomous daredevil pushes the limits of flight [Link]
24.06.2020	DailyMail , Drones all in a spin! AI algorithm enables quadcopters to perform acrobatic manoeuvres like power loops and barrel rolls autonomously [Link]
23.06.2020	Blick , Navigationsalgorithmus der Uni Zurich lehrt Drohnen Kunststuecklein [Link]
24.06.2020	Robohub , Drones learn acrobatics by themselves [Link]
24.06.2020	New Atlas , AI algorithm enables autonomous drones to do barrel rolls and flips [Link]
24.06.2020	InceptiveMind , A navigation algorithm enables drones to learn challenging acrobatic maneuvers [Link]
17.06.2020	DroneDj , Drones trained to do acrobatics thanks to artificial intelligence [Link]
27.03.2019	The New York Times A.I. Is Flying Drones (Very, Very Slowly) [Link]
27.06.2018	WIRED , Drones Just Learned to Fly Solo, Racers May Soon Meet Their Match [Link]
14.02.2018	La Repubblica , Tra alberi e palazzi ora il drone fa lo slalom [Link]
26.01.2018	Drone Life , DroNet Algorithm Learns From Traffic to Navigate City Streets [Link]
26.01.2018	The Robot Report , DroNet Teaches Drones to Autonomously Navigate Cities [Link]
26.01.2018	ZDNet , Autonomous high flying drones learn to navigate by watching traffic below [Link]
26.01.2018	Blick , Zürcher Algorithmus lenkt Drohnen sicher durch die Stadt [Link]
26.01.2018	MIT Tech Review , This drone learned to fly through streets by studying driverless-car data [Link]
25.01.2018	IEEE Spectrum , AI-Powered Drone Mimics Cars and Bikes to Navigate Through City Streets [Link]
24.01.2018	Tages Anzeiger , Diese Drohne lernt durch Imitation [Link]
24.01.2018	NZZ , So kommen Drohnen sicher durch die Stadt [Link]
24.01.2018	Digital Trends , The DroNet algorithm teaches drones to navigate city streets like cars [Link]
23.01.2018	Phys.org , Drones learn to navigate autonomously by imitating cars and bicycles [Link]
23.01.2018	Science Daily , Drones learn to navigate autonomously by imitating cars and bicycles [Link]
23.01.2018	Alpha Galileo , Drones learn to navigate autonomously by imitating cars and bicycles [Link]
23.01.2018	ORF Science , So kommen Drohnen sicher durch die Stadt [Link]
23.01.2018	Spektrum.de , Drohne lernt von Fahrradfahrern [Link]
23.01.2018	Blick am Abend , Sicher durch die Stadt [Link]
23.01.2018	20 Minuten , Uni macht Drohnenflüge sicherer [Link]

Advising

The parentheses indicate the student's current affiliation.

PhD Students

2024-	Chunwei Xing
2025-	Wei Wang (w/ Rahul Mangharam)
2026-	Vineet Pasumarti (w/ Vijay Kumar)
2026-	Orestis Konstantaropoulos

(Alumni) Master Students

2026	Margaret Mao (Foundation Robotics), Luyang Hu (Origami Robotics)
2024	Dylan Goetting (Skild AI)
2022	March Rauch (Daedalean AI)
2021	Nina Wiedermann (PhD, ETH Zurich)
2021	Simone Arreghini (PhD, IDSIA USI-SUPSI)
2021	Mario Bonsembiante (Amazon)
2021	Lorenzo Ferrini (Seervision AI)
2020	Alessandro Saviolo (PhD, New York University)
2020	Francesco Milano (PhD, ETH Zurich)
2019	Mattia Segu (PhD, ETH Zurich)
2019	Daniel Mouritzen (Torso Electronics)
2018	Simon Muntwiler (PhD, ETH Zurich)
2018	Moritz Zimmermann (Scandit)
2018	Bojana Nenezic (Danfoss)
2017	Yawei Ye (Waymo Research)

Visiting Students

Parenthesis indicate the student's home institution.

2026	Rama El-khawaldeh (University of British Columbia)
2026	Mattia Penzotti (Scuola Superiore Sant'Anna, Italy)
2025	Lorenzo Bianchi (Tor Vergata, Italy)

Professional Service

Organizer/Co-Organizer

2023	ICRA Workshop <i>Pre-training for Robotics</i> , London, UK. [Link]
2022	IROS Competition <i>Safe Robot Learning</i> , Tokyo, Japan. [Link]
2022	ICRA Competition <i>DodgeDrone: Vision-Based Agile Flight</i> , Philadelphia, USA. [Link]
2021	ICRA Workshop <i>Perception and Action in Dynamic Environments</i> , Online, [Link]
2020	AAAI Spring Symposium <i>ML for Mobile Robot Navigation in the Wild</i> , Palo Alto, California

Technical Reviewer

I reviewed each year for the following conferences and journals since 2018:	
Journals	IEEE Transactions on Robotics (T-RO) ● IEEE Robotics and Automation Letters (RA-L) ● Science Robotics ● Transactions on Pattern Analysis and Machine Intelligence (T-PAMI) ● The International Journal of Robotics Research (IJRR) ● Computer Graphics Forum ● Nature
Conferences	<i>Robotics</i> : Robotics: Science and Systems (RSS) ● Conference on Robotics Learning (CORL) ● International Conference on Robotics and Automation (ICRA) ● International Conference on Intelligent Robots and Systems (IROS) <i>Computer Vision</i> : Computer Vision and Pattern Recognition (CVPR) ● International/European Conference on Computer Vision (ICCV/ECCV) <i>Machine Learning</i> : Conference on Neural Information Processing Systems (NeurIPS) ● International Conference on Machine Learning (ICML) ● International Conference on Learning Representations (ICLR)
Books	Foundations and Trends in Robotics (Now Publishers) Panelist (2026)

Service

I have taken the role of area chair for:	
2026	Conference on Robot Learning (CORL)
2025	Robotics: Science and Systems (RSS)
2025	International Conference on Robotics and Automation (ICRA)
2024	Intelligent Robots and Systems (IROS)

Invited Speaker

Mar 26	Conference Keynote : Simulation: What makes us intelligent will make our robots intelligent, ICPRAM and ROBOVIS.
Nov 25	Department Colloquium : Simulation: What makes us intelligence will make our robots intelligent. University of Oregon AI Colloquium.
June 25	Workshop Keynote : 3D Multi-Sensory Perception, CVPR Workshop on Embodied Intelligence for Autonomous Systems on the Horizon.
May 25	Workshop Keynote : Towards a Theory of Sim-to-Real, ICRA Workshop on Data-Driven Control and Trajectory Optimization for Autonomy
Jan 25	Department Colloquium : Simulation: What made us intelligent will make our robots intelligent, ECE Department Seminar, Temple University.
Oct 24	Seminar : Simulation: What made us intelligent will make our robots intelligent, GRASP for Robotics Seminar, University of Pennsylvania.
June 24	Workshop Keynote : Towards Multi-Sensory World Models, Workshop on Understanding Higher-Level Intelligence from AI, Psychology, and Neuroscience Perspectives, The Simons Insitute.
May 24	Conference Keynote : Towards Multi-Sensory World Models, First International Conference on Neuro-symbolic Systems (NeuS).
April 24	Department Colloquium : Lessons learned from superhuman autonomous drone racing, DREAM Seminar, UC Berkeley.
Jan 24	Seminar : Lessons learned from superhuman autonomous drone racing, PRISMA Seminar, University of Naples Federico II.
Dec 23	Seminar : Learning vision-based pursuit-evasion policies, Multi-Agent Reinforcement Learning Seminar.
Oct 22	Workshop Keynote : Safe Robotics and the value of competitions for robotics, IROS.
May 22	Workshop Keynote : Workshop on Releasing Robots into the Wild, ICRA.

May 22	Workshop Keynote Aerial Robotics Workshop, ICRA.
May 22	Workshop Keynote Aerial Robotics Workshop, ICRA.
May 22	Workshop Keynote the SeasonDepth Prediction Challenge, ICRA.
Mar 22	Seminar AI in Robotics Seminar, University of Toronto [Youtube]
Feb 22	Department Colloquium Computer vision and robotics at an ELLIS Seminar in Turin
Nov 20	Seminar Autonomy Talks, ETH Zurich, [Youtube]
Nov 20	Workshop Keynote: UZH Machine Learning Workshop , [Link]
May 20	Workshop Keynote: Workshop on Perception, Action, and Learning, ICRA (with Davide Scaramuzza) [Youtube]
Apr 20	Workshop Keynote: Workshop Bridging AI and Cognitive Science (BAICS), ICLR
Apr 20	Seminar UZH Deep Learning Symposium, Zurich
Nov 19	Seminar Zurich Machine Learning Meetup

Publication List

For preprints, please check my google scholar page.

BOOKS

1. A. **Loquercio**, “Agile autonomy: Learning high-speed vision-based flight,” *Springer Tracts in Advanced Robotics*, 2023. DOI: [10.1007/978-3-031-27288-2](#)
2. A. Velasquez, V. Patel, A. **Loquercio**, and A. Fern, “Artificial intelligence: From simulation to reality,” *IEEE Wiley (in preparation)*, 2026

JOURNAL ARTICLES

1. S. Teetaert, W. Zhao, A. **Loquercio**, S. Zhou, L. Brunke, M. Schuck, W. Hönig, J. Panerati, and A. P. Schoellig, “Advancing reproducibility, benchmarks, and education with remote sim2real: Remote simulation to real robot hardware,” *IEEE Robotics and Automation Magazine*, vol. 32, no. 1, pp. 117–123, 2025. DOI: [10.1109/MRA.2025.3527291](#)
2. D. Zhang, A. **Loquercio**, J. Tang, T.-H. Wang, J. Malik, and M. W. Mueller, “A learning-based quadcopter controller with extreme adaptation,” *IEEE Transactions on Robotics*, 2025
3. D. Hanover, A. **Loquercio**, L. Bauersfeld, A. Romero, R. Penicka, Y. Song, G. Cioffi, E. Kaufmann, and D. Scaramuzza, “Autonomous drone racing: A survey,” *IEEE Transactions on Robotics*, 2024
4. E. Kaufmann, L. Bauersfeld, A. **Loquercio**, M. Müller, V. Koltun, and D. Scaramuzza, “Champion-level drone racing using deep reinforcement learning,” *Nature*, vol. 620, no. 7976, pp. 982–987, 2023
5. P. Foehn, E. Kaufmann, A. Romero, R. Penicka, S. Sun, L. Bauersfeld, T. Laengle, G. Cioffi, Y. Song, A. **Loquercio**, *et al.*, “Agilicious: Open-source and open-hardware agile quadrotor for vision-based flight,” *Science Robotics*, vol. 7, no. 67, 2022
6. C. Pfeiffer, S. Wengeler, A. **Loquercio**, and D. Scaramuzza, “Visual attention prediction improves performance of autonomous drone racing agents,” *Plos one*, vol. 17, no. 3, 2022

7. A. **Loquercio**, A. Saviolo, and D. Scaramuzza, “Autotune: Controller tuning for high-speed flight,” *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4432–4439, 2022
8. A. **Loquercio**, E. Kaufmann, R. Ranftl, M. Müller, V. Koltun, and D. Scaramuzza, “Learning high-speed flight in the wild,” *Science Robotics*, vol. 6, no. 59, 2021
9. A. **Loquercio**, A. Dosovitskiy, and D. Scaramuzza, “Learning depth with very sparse supervision,” *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5542–5549, 2020
10. A. **Loquercio**, M. Segu, and D. Scaramuzza, “A general framework for uncertainty estimation in deep learning,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3153–3160, 2020
11. A. **Loquercio**, E. Kaufmann, R. Ranftl, A. Dosovitskiy, V. Koltun, and D. Scaramuzza, “Deep drone racing: From simulation to reality with domain randomization,” *IEEE Transactions on Robotics*, vol. 36, no. 1, pp. 1–14, 2019
12. D. Palossi, A. **Loquercio**, F. Conti, E. Flamand, D. Scaramuzza, and L. Benini, “A 64-mw dnn-based visual navigation engine for autonomous nano-drones,” *IEEE Internet of Things Journal*, vol. 6, no. 5, pp. 8357–8371, 2019
13. A. **Loquercio**, A. I. Maqueda, C. R. Del-Blanco, and D. Scaramuzza, “Dronet: Learning to fly by driving,” *IEEE Robotics and Automation Letters*, vol. 3, no. 2, pp. 1088–1095, 2018

CONFERENCE ARTICLES

1. W. M. Mao, J. Ng, L. Hu, D. Gehrig, and A. **Loquercio**, “Roshi: A versatile robot-oriented suit for human data in-the-wild,” *International Conference on Intelligent Robots and Systems (IROS)*, 2026
2. V. Pasumarti, L. Bianchi, and A. **Loquercio**, “Agile flight emerges from multi-agent competitive racing,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026
3. H. G. Singh, P. Abbeel, J. Malik, and A. **Loquercio**, “Deep sensorimotor control by imitating predictive models of human motion,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026
4. D. Goetting, H. G. Singh, and A. **Loquercio**, “End-to-end navigation with vision language models: Transforming spatial reasoning into question-answering,” *2nd International Conference on Neuro-symbolic Systems (NeuS)*, 2025
5. Y. Dou, W. Oh, Y. Luo, A. **Loquercio**, and A. Owens, “Hearing hands: Generating sounds from physical interactions in 3d scenes,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025
6. H. G. Singh, A. **Loquercio**, C. Sferrazza, J. Wu, H. Qi, P. Abbeel, and J. Malik, “Hand-object interaction pretraining from videos,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025

7. H. Huang, A. **Loquercio**, A. Kumar, N. Thakkar, K. Goldberg, and J. Malik, “Manipulator as a tail: Promoting dynamic stability for legged locomotion,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 9712–9719. DOI: [10.1109/ICRA57147.2024.10610049](https://doi.org/10.1109/ICRA57147.2024.10610049)
8. H. Huang, S. Sharma, A. **Loquercio**, A. Angelopoulos, K. Goldberg, and J. Malik, “Conformal policy learning for sensorimotor control under distribution shifts,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 16 285–16 291. DOI: [10.1109/ICRA57147.2024.10610223](https://doi.org/10.1109/ICRA57147.2024.10610223)
9. A. Bar, A. Bakhtiar, D. Tran, A. **Loquercio**, J. Rajasegaran, Y. LeCun, A. Globerson, and T. Darrell, “Egopet: Egomotion and interaction data from an animal’s perspective,” in *European Conference on Computer Vision (ECCV)*, 2024
10. Y. Dou, F. Yang, Y. Liu, A. **Loquercio**, and A. Owens, “Tactile-augmented radiance fields,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 26 529–26 539
11. A. Bajcsy, A. **Loquercio**, A. Kumar, and J. Malik, “Learning vision-based pursuit-evasion robot policies,” *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024
12. H. Huang, A. **Loquercio**, A. Kumar, N. Thakkar, K. Goldberg, and J. Malik, “More than an arm: Using a manipulator as a tail for enhanced stability in legged locomotion,” *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024
13. H. Huang, S. Sharma, A. **Loquercio**, A. Angelopoulos, K. Goldberg, and J. Malik, “Conformal policy learning for sensorimotor control under distribution shifts,” *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024
14. A. **Loquercio**, A. Kumar, and J. Malik, “Learning visual locomotion with cross-modal supervision,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2023, pp. 7295–7302
15. N. Wiedemann, V. Wüest, A. **Loquercio**, M. Müller, D. Floreano, and D. Scaramuzza, “Training efficient controllers via analytic policy gradient,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2023, pp. 1349–1356
16. D. Zhang, A. **Loquercio**, X. Wu, A. Kumar, J. Malik, and M. W. Mueller, “Learning a single near-hover position controller for vastly different quadcopters,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2023, pp. 1263–1269
17. A. **Loquercio** and D. Scaramuzza, “Agile autonomy: High-speed flight with on-board sensing and computation,” in *Conference on Robotics and Intelligent Machines (I-RIM)*, 2020
18. F. Milano, A. **Loquercio**, A. Rosinol, D. Scaramuzza, and L. Carlone, “Primal-dual mesh convolutional neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020
19. Y. Song, S. Naji, E. Kaufmann, A. **Loquercio**, and D. Scaramuzza, “Flightmare: A flexible quadrotor simulator,” in *Conference on Robot Learning*, 2020

20. N. Messikommer, D. Gehrig, A. **Loquercio**, and D. Scaramuzza, “Event-based asynchronous sparse convolutional networks,” in *European Conference on Computer Vision (ECCV)*, 2020
21. E. Kaufmann*, A. **Loquercio***, R. Ranftl, M. Müller, V. Koltun, and D. Scaramuzza, “Deep drone acrobatics,” in *Robotics, Science, and Systems (RSS)*, 2020
22. D. Gehrig, A. **Loquercio**, K. G. Derpanis, and D. Scaramuzza, “End-to-end learning of representations for asynchronous event-based data,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019
23. Y. Yang*, A. **Loquercio***, D. Scaramuzza, and S. Soatto, “Unsupervised moving object detection via contextual information separation,” in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2019, pp. 879–888
24. E. Kaufmann*, A. **Loquercio***, R. Ranftl, A. Dosovitskiy, V. Koltun, and D. Scaramuzza, “Deep drone racing: Learning agile flight in dynamic environments,” in *Conference on Robotic Learning (CoRL)*, 2018
25. A. I. Maqueda, A. **Loquercio**, G. Gallego, N. García, and D. Scaramuzza, “Event-based vision meets deep learning on steering prediction for self-driving cars,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018
26. Y. Ye, T. Cieslewski, A. **Loquercio**, and D. Scaramuzza, “Place recognition in semi-dense maps: Geometric and learning-based approaches,” in *British Machine Vision Conference (BMVC)*, 2017
27. A. **Loquercio**, M. Dymczyk, B. Zeisl, S. Lynen, I. Gilitschenski, and R. Siegwart, “Efficient descriptor learning for large scale localization,” in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, 2017, pp. 3170–3177